simple_led_routed

Design Report

kicad-tools 0.23.0

Rev 1 | 2026-10-18 | jlcpcb

Board Summary

Property	Value
Layers	2 copper (F.Cu, B.Cu)
Footprints	3 (1 SMD, 2 THT, 0 other)
Nets	3
Traces	2 segments
Vias	0
Board Size	25.0 x 20.0 mm

Design Overview

Theory of Operation

Simple LED - Hello World

Minimal LED circuit

5V input, 330 ohm resistor, LED

Power Architecture

Power Rails: GND, VCC

Assembly Notes

1 polarized component

- **Polarized components**: 1 – check orientation markings

ERC Status

Metric	Count
Errors	0
Warnings	0

Status: PASS

Schematic Overview

Schematic: simple_led

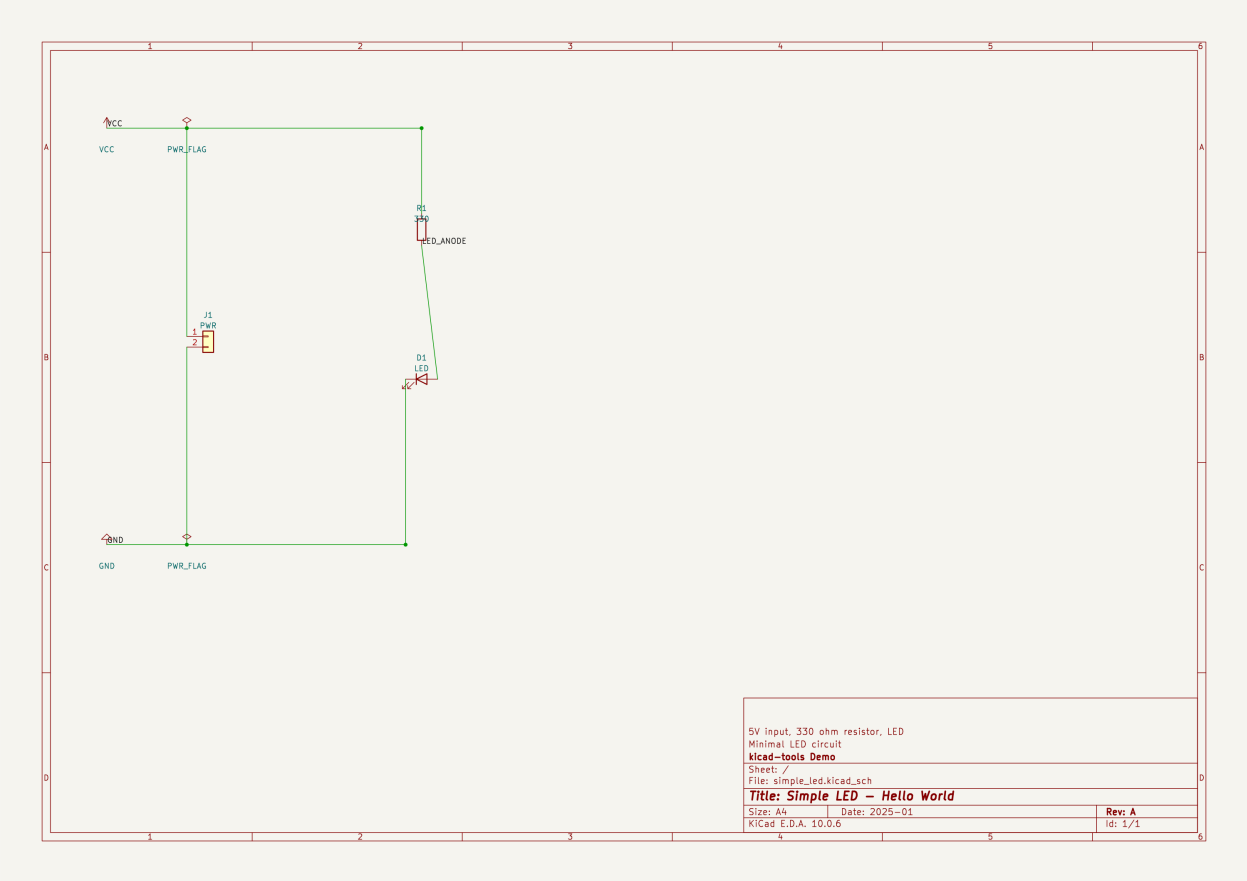


Figure 1: Schematic: simple_led

PCB Layout

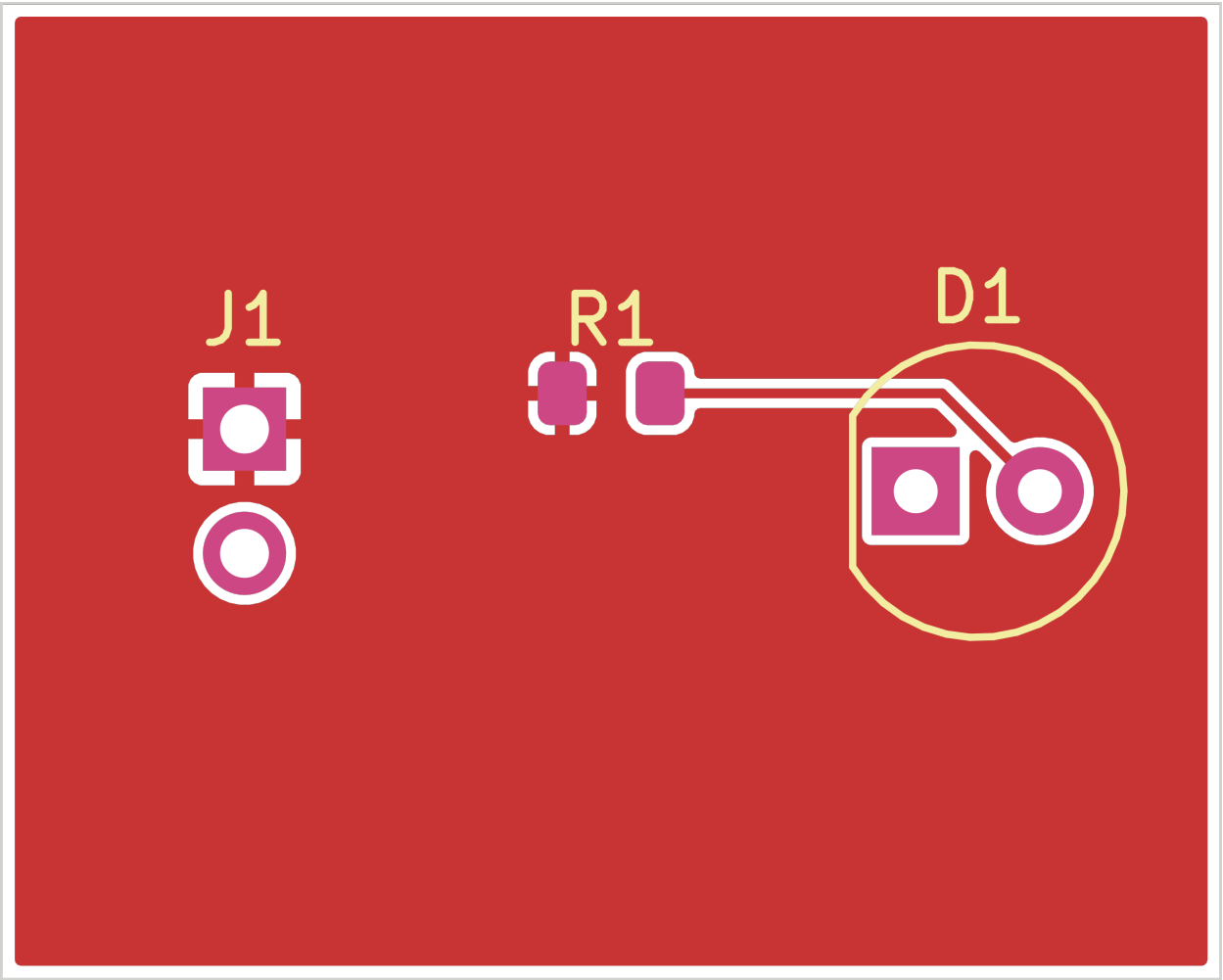


Figure 2: PCB Front

Copper

Assembly



Figure 3: PCB Back

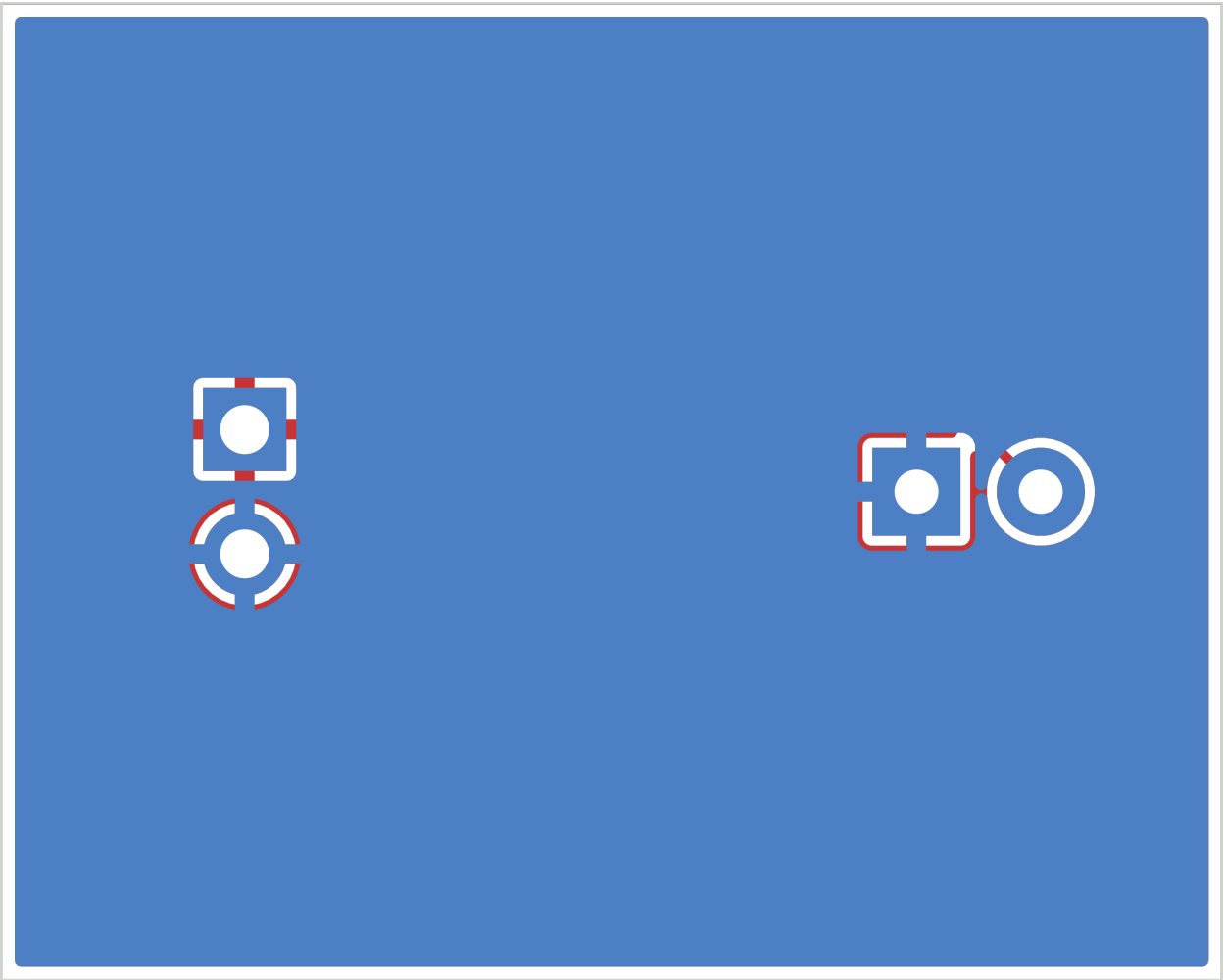


Figure 4: PCB Copper

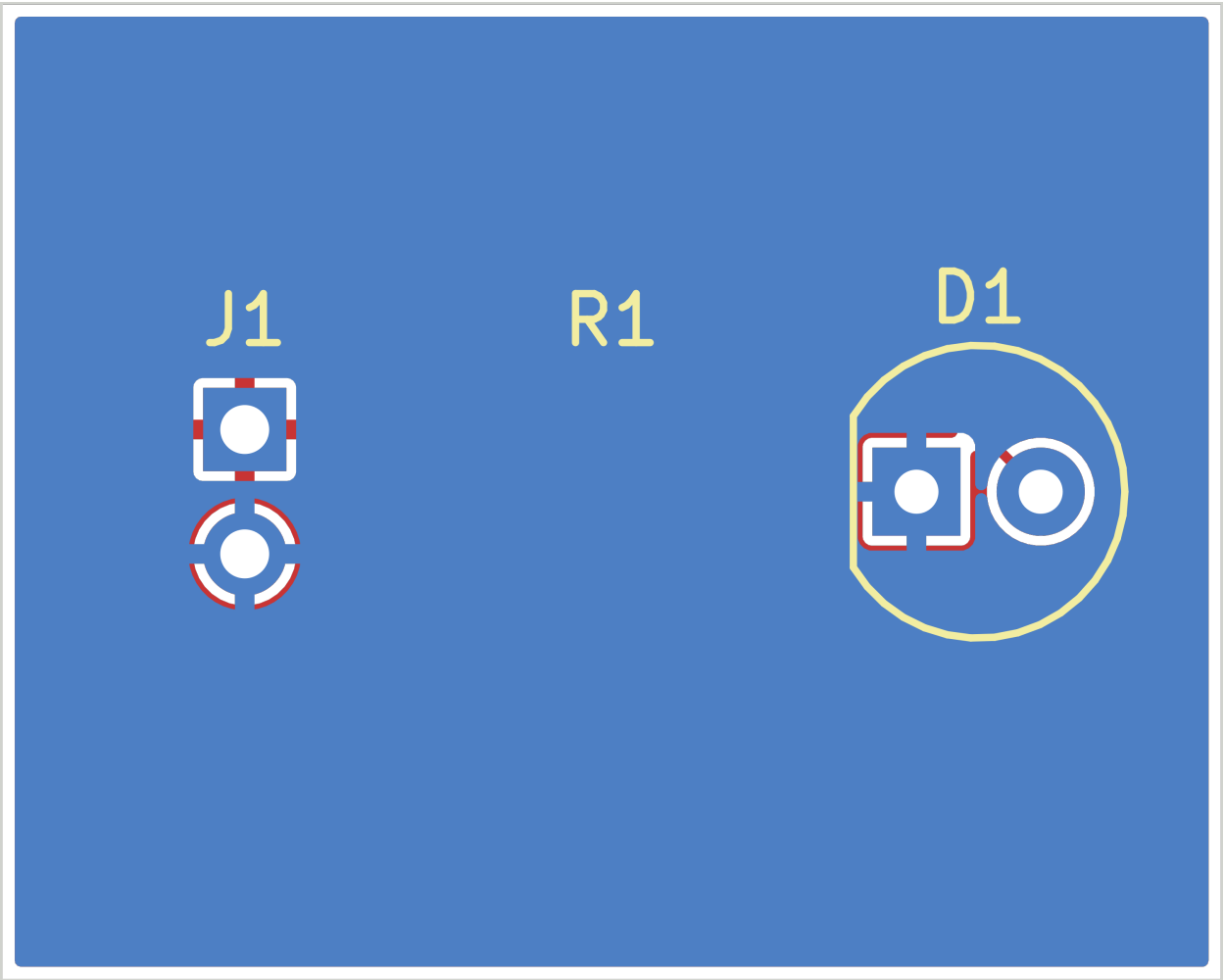


Figure 5: Assembly

Copper Layers

F.Cu

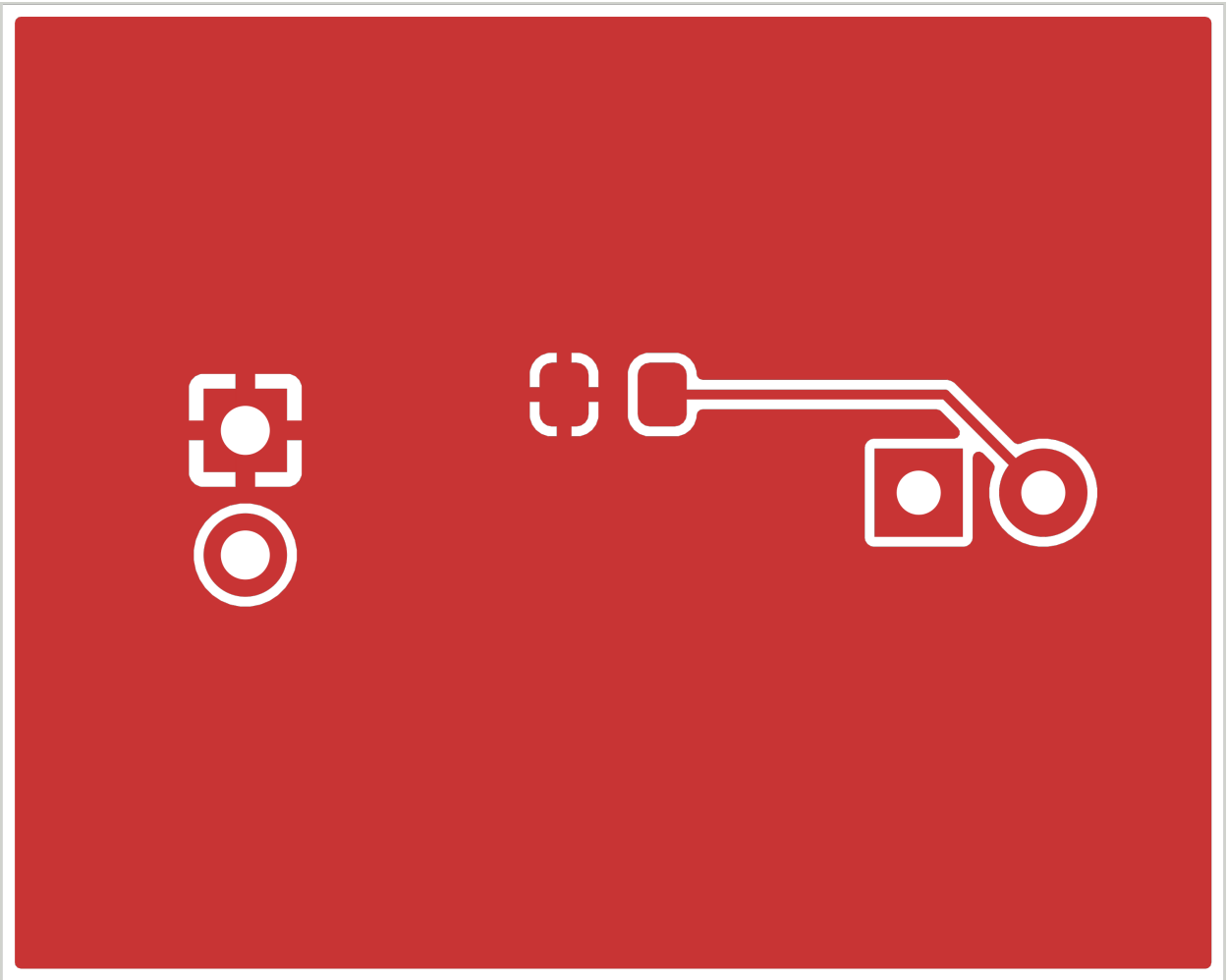


Figure 6: F.Cu

B.Cu

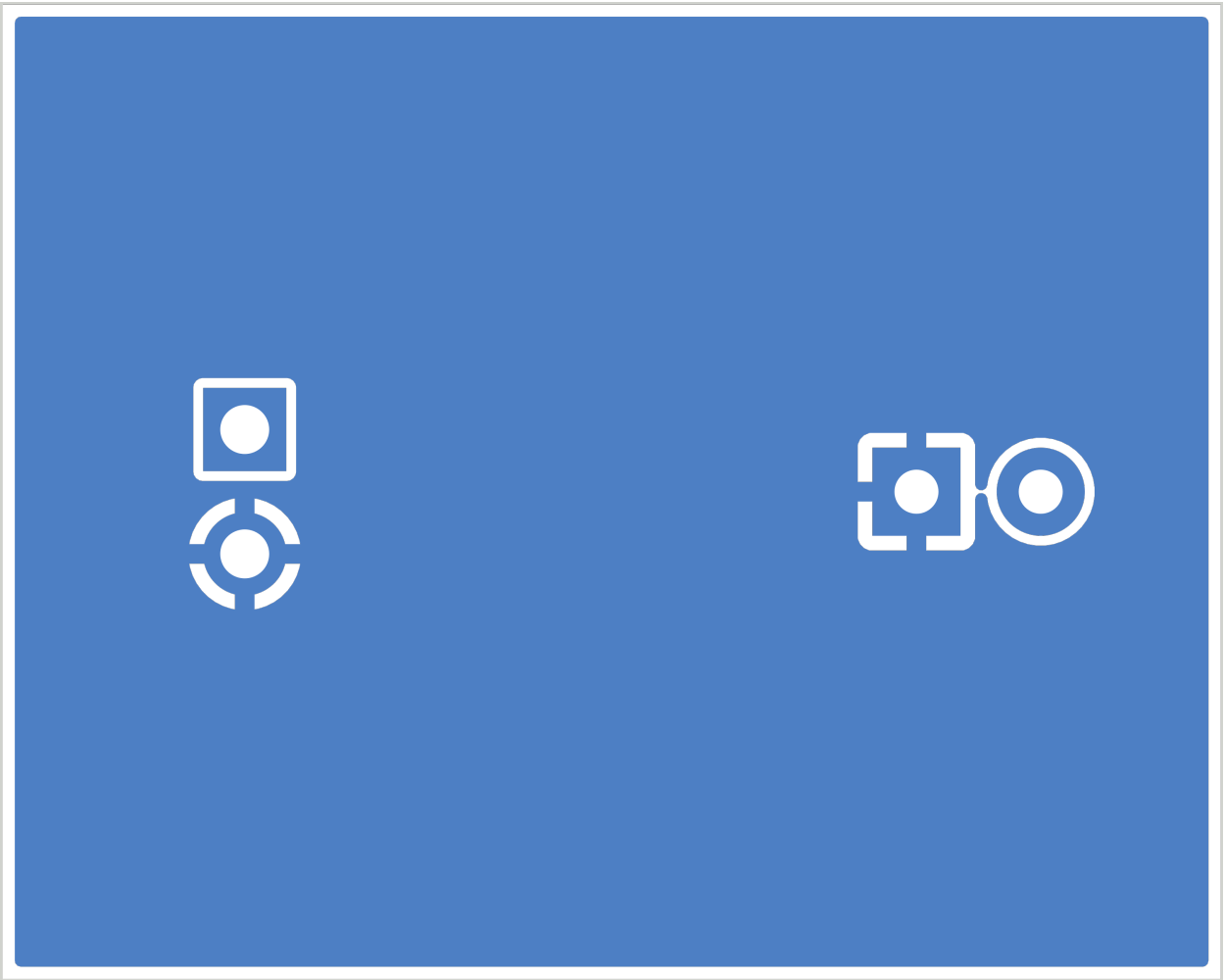


Figure 7: B.Cu

Bill of Materials

Value	Package	Qty	References
LED	LED_D5.0mm	1	D1
PWR	PinHeader_1x02_P2.54mm_Vertical	1	J1
330	R_0805_2012Metric	1	R1

DRC Status (`--mfr jlcpcb` rules)

Metric	Count
Errors	0
Errors: kct (`--mfr jlcpcb` rules)	0
Errors: kicad-cli (geometric)	0
Warnings	0
Blocking	0

Status: PASS

Manufacturing Readiness

Verdict: READY

Action Items

- [**OPTIONAL**] Refill zones to verify 2 zone-connected nets, then re-run kct audit

Routing Status

Metric	Value
Signal Net Completion	100.0% (1/1)
Overall Completion	100.0%
Complete Nets	3 / 3
Zone-Connected Nets	2
Incomplete Nets	0
Unconnected Pads	0

Zone-Connected Nets

- GND
- VCC

Cost Estimate

Metric	Per Board (estimated)
PCB Fabrication	~0.5 USD
Components (estimated)	~0.12 USD
Assembly (estimated)	~1.92 USD
Total (estimated)	**~2.55 USD**
Batch Quantity	5
Batch Total (estimated)	~12.73 USD